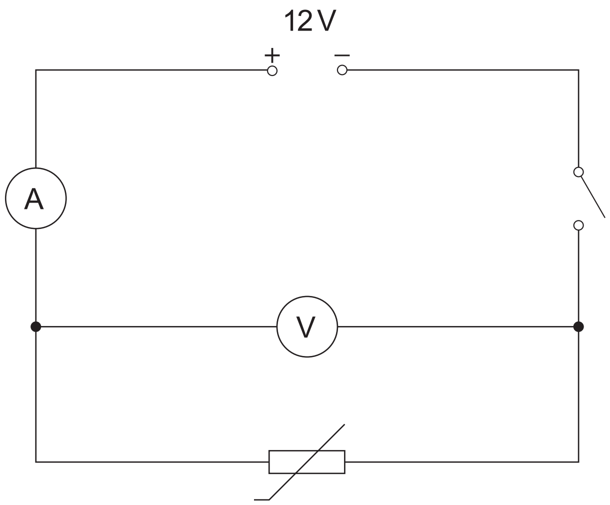
4. A thermistor is a type of resistor whose resistance depends on temperature. Thermistors are often used as temperature sensors in applications where we want to monitor or control temperature. A group of students immerse a thermistor and a thermometer in water in order to determine the resistance at different temperatures. The circuit they use is shown below.



Their results are given below.

Temperature (°C)	Current (mA)	Voltage (V)	Resistance (kΩ)
20	0.97	12.0	12.4
40	2.22	12.0	5.41
60	4.80	12.0	2.50
80	9.23	12.0	1.30
100	17.14	12.0	0.70

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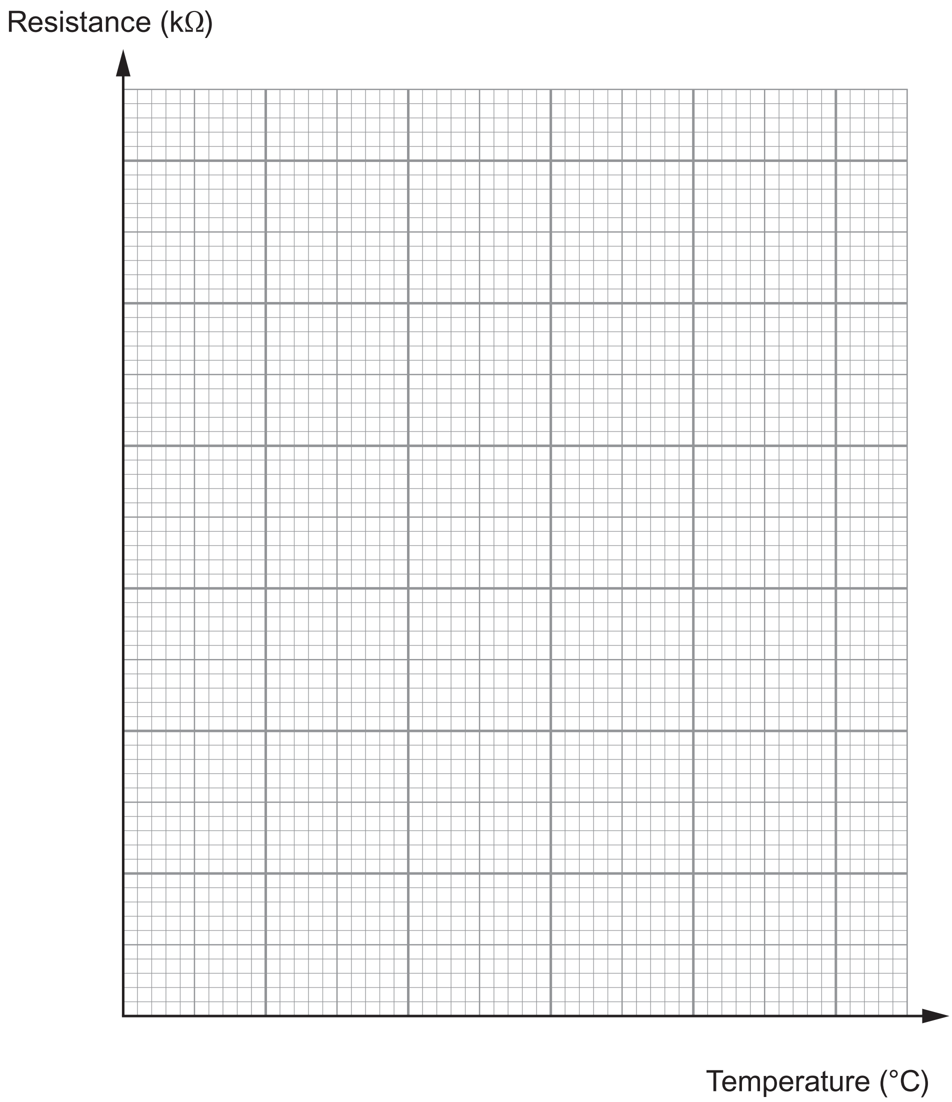
(a) (i) Plot the data on the grid below and draw a suitable line. [4]



(ii) Describe the relationship between temperature and resistance. [2]

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(b) (i) A resistor of resistance 5 kΩ is now added to the original circuit, in parallel with the thermistor which is at a temperature of 30 °C. The same 12 V power supply is used. Use your graph and equations from page 2 to calculate the total current in the circuit. [5]

Current = ... A

(ii) Kim suggests that the current through both the resistor and the thermistor will decrease as the thermistor cools down. Explain whether or not she is correct. [3]

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